

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data to Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a PySpark job to load customer and order data from CSV files into Delta tables.

Target Cluster Configuration:

- Cluster Name: Data Engineering Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Customer data CSV file: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata``
- Order data CSV file: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``

Target Data Details:

- Customer Delta table: ``customer`` in ``gen_ai_poc_databrickscoe.sdlc_wizard``
- Order Delta table: ``order`` in ``gen_ai_poc_databrickscoe.sdlc_wizard``

Job Flow / Pipeline Stages:

- Read customer data from CSV file
- Load customer data into Delta table
- Read order data from CSV file
- Load order data into Delta table

Data Transformations / Business Logic:

- None

Error Handling and Logging:

- Log errors to a file in ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/logs``

TR-DLT-002: Transform Order Data by Adding TotalAmount Column

Related Functional Requirement(s): FR-DLT-002

Objective: Implement a PySpark job to add a new column "TotalAmount" to the order Delta table.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- Order Delta table: `order` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`

Target Data Details:

- Transformed Order Delta table: `order` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`

Job Flow / Pipeline Stages:

- Read order data from Delta table
- Add new column "TotalAmount" to order data
- Write transformed data back to Delta table

Data Transformations / Business Logic:

- Calculate "TotalAmount" as the product of "PricePerUnit" and "Qty" columns

Error Handling and Logging:

- Log errors to a file in `/Volumes/gen\_ai\_poc\_databrickscoe/sdmc\_wizard/logs`

TR-DLT-003: Clean Customer and Order Data

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a PySpark job to remove null and duplicate records from customer and order Delta tables.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- Customer Delta table: `customer` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`
- Order Delta table: `order` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`

Target Data Details:

- Cleaned Customer Delta table: `customer` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`
- Cleaned Order Delta table: `order` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`

Job Flow / Pipeline Stages:

- Read customer data from Delta table

- Remove null and duplicate records from customer data
- Write cleaned data back to Delta table
- Read order data from Delta table
- Remove null and duplicate records from order data
- Write cleaned data back to Delta table

Data Transformations / Business Logic:

- Remove null records
- Remove duplicate records

Error Handling and Logging:

- Log errors to a file in ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/logs``

TR-DLT-004: Create ordersummary Table and Load Joined Data

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a PySpark job to create an "ordersummary" table and load joined data from customer and order Delta tables.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- Customer Delta table: ``customer`` in ``gen_ai_poc_databrickscoe.sdlc_wizard``
- Order Delta table: ``order`` in ``gen_ai_poc_databrickscoe.sdlc_wizard``

Target Data Details:

- ordersummary table: ``ordersummary`` in ``gen_ai_poc_databrickscoe.sdlc_wizard``

Job Flow / Pipeline Stages:

- Create "ordersummary" table if it does not exist
- Join customer and order data on "CustId" field
- Load joined data into "ordersummary" table as an SCD type 2 table

Data Transformations / Business Logic:

- Join customer and order data
- Implement SCD type 2 logic

Error Handling and Logging:

- Log errors to a file in ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/logs``

TR-DLT-005: Update ordersummary Table on Customer Data Change

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a PySpark job to update the "ordersummary" SCD type 2 table whenever there is a change in the customer table.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- Customer Delta table: `customer` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`

Target Data Details:

- ordersummary table: `ordersummary` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`

Job Flow / Pipeline Stages:

- Detect changes in customer data
- Update "ordersummary" table by making old records inactive and new records active
- Update "StartDate" and "EndDate" columns accordingly to maintain history

Data Transformations / Business Logic:

- Implement SCD type 2 logic

Error Handling and Logging:

- Log errors to a file in `/Volumes/gen\_ai\_poc\_databrickscoe/sdmc\_wizard/logs`

TR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data

Related Functional Requirement(s): FR-DLT-006

Objective: Implement a PySpark job to create a "customeraggregatespend" table and load aggregated data from "ordersummary" table.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- ordersummary table: `ordersummary` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`

Target Data Details:

- customeraggregatespend table: `customeraggregatespend` in `gen\_ai\_poc\_databrickscoe.sdmc\_wizard`

Job Flow / Pipeline Stages:

- Create "customeraggregatespend" table if it does not exist
- Aggregate "TotalAmount" column from "ordersummary" table by "Name" and "Date" columns
- Load aggregated data into "customeraggregatespend" table

Data Transformations / Business Logic:

- Aggregate data by "Name" and "Date" columns

Error Handling and Logging:

- Log errors to a file in `Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/logs`

TR-DLT-007: Implement Delta Live Tables**Related Functional Requirement(s): FR-DLT-007**

Objective: Implement Delta Live Tables for the above-mentioned functionality.

Target Cluster Configuration:

- Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Implement Delta Live Tables for loading customer and order data
- Implement Delta Live Tables for transforming order data
- Implement Delta Live Tables for cleaning customer and order data
- Implement Delta Live Tables for creating and loading "ordersummary" table
- Implement Delta Live Tables for updating "ordersummary" table on customer data change
- Implement Delta Live Tables for creating and loading "customeraggregatespend" table

Data Transformations / Business Logic:

- Same as above-mentioned TRs

Error Handling and Logging:

- Log errors to a file in `Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/logs`